

K L UNIVERSITY
FRESHMAN ENGINEERING DEPARTMENT

A Project Based Lab Report

On

Smart Finance Manager & Personal Expense Intelligenece System (Sfm-Pes)

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CERTIFICATE

This is to certify that the project based laboratory report entitled “MOVIE TICKET BOOKING” submitted by **ROHINI KALYANI, VENNELA, GOPAL AJAY KUMAR, PHANIDATTA**, bearing id no. **2500030783, 2500030822, 2500030718, 2500030632** to the **Department of COMPUTER SCIENCE ENGINEERING**, KL University in partial fulfillment of the requirements for the completion of a project based Laboratory in “Computational Thinking for Problem Solving (24SC1102)” course in I B Tech I Semester, is a Bonafede record of the work carried out by him/her under my supervision during the academic year 2024 – 2025.

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ABSTRACT

The **Smart Finance Manager & Personal Expense Intelligence System (SFM-PEIS)** is a Python-based application designed to help individuals effectively manage their personal finances. It provides a structured platform for recording daily income and expenses, categorizing transactions, and maintaining a clear overview of financial activities. The system simplifies complex financial tracking by offering an easy-to-use interface, enabling users to monitor their spending habits and maintain accurate financial records.

The application utilizes Python's data processing and analytical capabilities to evaluate user spending patterns and generate meaningful insights. It allows users to set budget limits for different categories such as food, travel, shopping, and utilities, and provides alerts when expenses exceed these limits. Additionally, the system generates detailed reports and visualizations, including charts and summaries, which help users better understand their financial behavior and make informed decisions. Basic predictive analysis is also incorporated to estimate future expenses based on past data. By providing intelligent insights and organized financial data, the system supports better money management and long-term financial

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INTRODUCTION

Managing personal finances is important but often difficult due to increasing daily expenses and lack of proper tracking methods. The *Smart Finance Manager & Personal Expense Intelligence System (SFM-PEIS)* is designed to provide a simple and efficient way to record and manage financial activities. This Python-based application helps users track their income and expenses in an organized managed. The system not only records transactions but also analyzes spending patterns. It allows users to categorize expenses, set budget limits, and receive alerts when they exceed their budget. This helps users control unnecessary spending and develop better financial habits. . It aims to provide a clear overview of a user's financial status, making it easier to track spending habits and manage money in a more organized way.

Additionally, the application provides visual reports and basic predictions to support better decision-making. By using charts and summaries, users can understand their financial trends easily. Overall, this project shows how technology can be used to improve personal financial management and planning.

AIM

The main aim of the *Smart Finance Manager & Personal Expense Intelligence System (SFM-PEIS)* is to develop a user-friendly and efficient application that helps in manage their personal finances effectively. The system is designed to simplify the process of recording daily income and expenses, categorizing financial transactions, and maintaining accurate financial records. It aims to provide a clear overview of a user's financial status, making it easier to track spending habits and manage money in a more organized way.

Another key aim of the project is to enhance financial awareness and decision-making by analyzing spending patterns and providing useful insights. The system helps users set budget limits, monitor their expenses, and receive alerts when they exceed their planned budget. Additionally, it aims to present financial data through visual reports and basic predictions, enabling users to understand trends and plan their future finances more effectively. Overall, the project focuses on promoting better financial discipline and smarter money management using technology.

ADVANTAGES

- * Helps users record income and expenses systematically, reducing confusion and manual errors.
- * Allows users to set budget limits and control unnecessary spending effectively.
- * Provides insights into spending patterns, helping users make smarter financial decisions.
- * Simple interface and automation save time compared to traditional manual methods.

DISADVANTAGES

- * May not include advanced financial tools like investment tracking or tax management.
- * Accuracy depends on users entering correct and regular data.
- * Financial data may be vulnerable if proper security measures are not implemented.
- * Some users may find it difficult to use or maintain without basic technical skills.

FUTURE IMPLEMENTATION

Future versions may also use AI and machine learning to provide smarter financial predictions and personalized saving suggestions.

SYSTEM REQUIREMENTS

➤ **SOFTWARE REQUIREMENTS:**

The major software requirements of the project are as follows: Language: Turbo-C.

Operating system: Windows 10 or more.

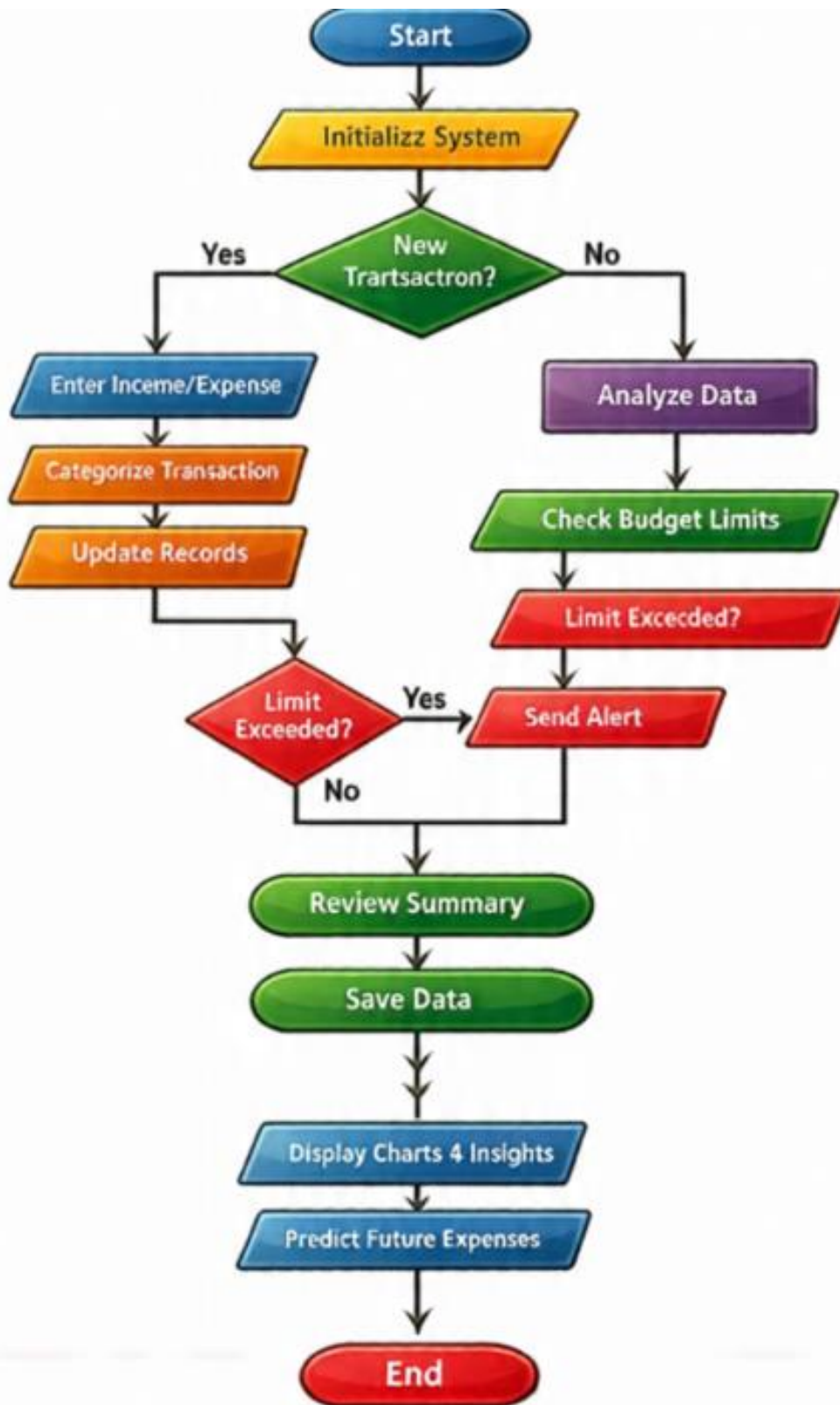
Technical requirements: Monitor, CPU, Keyboard, Mouse, etc...

➤ **HARDWARE REQUIREMENTS:**

The hardware requirements that map towards the software are as follows: RAM: 8 GB.

Processor: i5/i7.

FLOWCHART



ALGORITHM

Step 1: Start and Initialize System

Create data structures for income/expense records and set categories & budget.

Step 2: Display Menu

Show options: Add Income, Add Expense, View Transactions, Analyze Spending, Set Budget, Exit.

Step 3: Get User Choice

Accept input from the user.

Step 4: Add Income (if selected)

Enter and store income details.

Step 5: Add Expense (if selected)

Enter expense details, store them, and update total expenses.

Step 6: Check Budget Limit

Compare expenses with budget and display alert if exceeded.

Step 7: View & Analyze Data

Display transactions and generate spending analysis (category-wise summary).

Step 8: Repeat or Exit

Continue until user selects exit, then stop the program.

IMPLEMENTATION

```
import json

import datetime
FILE = "data.json"
def load():
    try:
        return json.load(open(FILE))
    except:
        return {"income": [], "expenses": [], "budget": {}}
def save(data):
    json.dump(data, open(FILE, "w"), indent=2)
    data = load()
    while True:
print("\n1.Income 2.Expense 3.Budget 4.View 5.Summary 6.Exit")
        ch = input("Choice: ")
        if ch == "1":
            amt = float(input("Amount: "))
            data["income"].append({
                "amt": amt,
                "date": str(datetime.date.today())
            })
        elif ch == "2":
            cat = input("Category: ")
            amt = float(input("Amount: "))
            data["expenses"].append({
                "amt": amt,
                "cat": cat,
                "date": str(datetime.date.today())
            })
        if cat in data["budget"] and amt > data["budget"][cat]:
            print("⚠ Budget Exceeded!")
```

```
        elif ch == "3":
            cat = input("Category: ")
            limit = float(input("Limit: "))
            data["budget"][cat] = limit
        elif ch == "4":
            print("\nIncome Records:")
            for i in data["income"]:
                print(i)
            print("\nExpense Records:")
            for e in data["expenses"]:
                print(e)
        elif ch == "5":
            inc = sum(i["amt"] for i in data["income"])
            exp = sum(e["amt"] for e in data["expenses"])
            print("\nTotal Income:", inc)
            print("Total Expense:", exp)
            print("Balance:", inc - exp)
        elif ch == "6":
            print("Exiting...")
            break
        else:
            print("Invalid Choice!")

save(data)
```

INTEGRATION AND SYSTEM TESTING OUTPUT AND SCREENSHOTS:

```
C:\Users\rohin\PyCharmMiscProject\.venv\Scripts\python.exe C:\Users\rohin\PyCharmMiscProject\pp.p
```

```
1.Income 2.Expense 3.Budget 4.View 5.Summary 6.Exit
```

```
Choice: 1
```

```
Amount: 100000
```

```
1.Income 2.Expense 3.Budget 4.View 5.Summary 6.Exit
```

```
Choice: 4
```

```
Income Records:
```

```
{'amt': 500.0, 'date': '2026-04-07'}
```

```
{'amt': 20000.0, 'date': '2026-04-07'}
```

```
{'amt': 100000.0, 'date': '2026-04-07'}
```

```
Expense Records:
```

```
{'amt': 400.0, 'cat': '5', 'date': '2026-04-07'}
```

```
{'amt': 600.0, 'cat': '5', 'date': '2026-04-07'}
```

```
{'amt': 10000.0, 'cat': '3', 'date': '2026-04-07'}
```

```
{'amt': 6000.0, 'cat': '3', 'date': '2026-04-07'}
```

```
C:\Users\rohin\PyCharmMiscProject\.venv\Scripts\python.exe C:\Users\rohin\PyCharmMiscProject\pp.py
```

```
1.Income 2.Expense 3.Budget 4.View 5.Summary 6.Exit
```

```
Choice: 3
```

```
Category: 5
```

```
Limit: 10000
```

```
1.Income 2.Expense 3.Budget 4.View 5.Summary 6.Exit
```

```
Choice: 5
```

```
Total Income: 120500.0
```

```
Total Expense: 17000.0
```

```
Balance: 103500.0
```

```
1.Income 2.Expense 3.Budget 4.View 5.Summary 6.Exit
```

```
Choice: |
```

```
C:\Users\rohin\PyCharmMiscProject\.venv\Scripts\python.exe C:\Users\rohin\PyCharmMiscProject\pp.py
```

```
1.Income 2.Expense 3.Budget 4.View 5.Summary 6.Exit
```

```
Choice: 1
```

```
Amount: 100000
```

```
1.Income 2.Expense 3.Budget 4.View 5.Summary 6.Exit
```

```
Choice: 2
```

```
Category: 3
```

```
Amount: 20000
```

```
1.Income 2.Expense 3.Budget 4.View 5.Summary 6.Exit
```

```
Choice: 5
```

```
Total Income: 220500.0
```

```
Total Expense: 37000.0
```

```
Balance: 183500.0
```

```
1.Income 2.Expense 3.Budget 4.View 5.Summary 6.Exit
```

```
Choice: |
```

CONCLUSION

This project presents a simple and effective Personal Finance Tracker developed using Python, which enables users to manage their income, expenses, and budgets in an organized manner. By storing data in a JSON file, the system ensures persistence and easy retrieval of financial records. The menu-driven interface allows users to perform essential operations such as adding transactions, setting budget limits, viewing records, and generating summaries with minimal complexity. Additionally, the budget alert feature helps users maintain financial discipline by notifying them when spending exceeds predefined limits. Overall, the project demonstrates how basic programming concepts can be applied to build a practical tool for improving personal financial awareness and decision-making.

Furthermore, the project is designed with simplicity and scalability in mind, making it suitable for beginners while also allowing room for future enhancements. Features such as categorizing expenses, visualizing spending patterns through graphs, or integrating with databases and mobile applications can be added to improve functionality. The use of Python ensures readability and ease of maintenance, encouraging users to modify and expand the system according to their needs. This project not only strengthens programming skills but also promotes responsible financial habits, making it a valuable learning experience with real-world applications.

